



# ***CharAnalysis 2.0: June 2026 Update***

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**SNI warning · Chronological uncertainty ensemble**

GitHub “dev” branch:

- NEWS [https://github.com/phiguera/CharAnalysis/blob/dev/CharAnalysis\\_2\\_0\\_R/NEWS.md](https://github.com/phiguera/CharAnalysis/blob/dev/CharAnalysis_2_0_R/NEWS.md)
- Chronological Uncertainty Vignette:  
[https://github.com/phiguera/CharAnalysis/blob/dev/CharAnalysis\\_2\\_0\\_R/vignettes/chronological\\_uncertainty\\_draft.md](https://github.com/phiguera/CharAnalysis/blob/dev/CharAnalysis_2_0_R/vignettes/chronological_uncertainty_draft.md)

Phil Higuera — 25 June, 2026





# Today's updates

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## 1 SNI warning

When SNI falls below 3.0, *CharAnalysis* now flags it explicitly in the console.

The vignette's recommended workflow was rewritten to position SNI evaluation as Step 2, before any fire-history interpretation.

## 2 Chronological uncertainty

New ensemble pipeline: runs *CharAnalysis* 1,000× across MCMC chronologies, propagating age uncertainty through all analytical steps.

Reports per-peak:

- Detection frequency
- Timing uncertainty (95% CI)
- Ensemble-only candidate peaks.

Examples from CO, CH10, and SI17.



# 1. SNI warning:

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## Before

SNI was computed and plotted (Fig 4), but no message appeared if it fell below the threshold of 3.0. Users could overlook low-SNI intervals.

## After

*CharAnalysis* now emits a styled red advisory at the end of all console output when  $SNI < 3.0$ .

**Local thresholds:** reports % below threshold

**Global threshold:** reports the scalar SNI value

R Console

### CAUTION

✗ 12.8% of samples have an  $SNI < 3.0$ , the minimum value recommended by Kelly et al. (2011) to identify records suitable for peak detection. **Carefully consider whether and how to interpret portions of this record with  $SNI < 3.0$ .**

i Run `'char_plot_sni(out)'` to visualize the SNI series and identify which portion(s) of the record fall below the threshold.



# Vignette: SNI evaluation before interpretation

**1 Background smoothing**  
Figs 1 & 3: check that  $C_{\text{background}}$  tracks low-frequency variation without absorbing peaks. Adjust smoothing window as needed.

**2 Evaluate signal quality (SNI)**  
Fig 4: inspect SNI before any fire-history results. If  $\text{SNI} < 3$  for substantial portions, peak detection in those intervals is not well justified.

**3 Threshold diagnostics**  
Fig 2: examine fit of noise model to  $C_{\text{peak}}$  distribution. Most useful when Fig 4 suggests instability.

**4 Analytical figures**  
Figs 6–9: zone comparisons, cumulative peaks, FRI distributions, fire history. Only after diagnostics are acceptable.



## 2. Chronological uncertainty

***CharAnalysis* alone does not propagate uncertainty from an age-depth model into peak detection** because it uses one age-depth model (e.g., median chronology).

Bayesian age-depth models (rbacon, MCAgeDepth) generate thousands of plausible chronologies via MCMC sampling. Each shifts sample ages and thus accumulation rates, CHAR, thresholds, and peak detection outcomes.

**This tool integrates chronological uncertainty to address three questions:**

1. How consistently is each peak identified?
2. What is the range of ages assigned to specific peaks?
3. Are there peaks common in the ensemble but absent from the main *CharAnalysis* run?

### **Most valuable for:**

- Identifying & timing individual peaks (e.g., for comparison to other events)
- Records with meaningful age-model uncertainty

### **Less critical for:**

- Fire-regime statistics (mFRI, fire frequency), because timing shifts among peaks tend to covary.



# Ensemble workflow: four steps

1

## Build age-depth model

*(external: rbacon /  
MCAgeDepth)*

Extract  $n = 1,000$  chronologies from the MCMC output, preserving temporal covariance.

Any age-depth tool works; rbacon is the primary target.

2

## Run the ensemble

*(char\_run\_ensemble.R)*

*CharAnalysis* pipeline runs once per chronology.

**Parallel processing:**  
~2 min for 1,000 iterations on a modern desktop (19 cores).

3

## Analyze ensemble results

*(run\_ensemble\_analysis.R)*

Use adaptive matching window for each reference peak.

### Reports:

- Detection frequency
- 95% CI on timing
- Ensemble-only peaks (10% threshold).

4

## Visualize

*(plot\_ensemble\_figure.R)*

### Four-panel figure:

- CHAR + benchmark
- Detection frequency + peak types
- SNI
- Age-depth uncertainty ribbon.

# Adaptive matching window

For each benchmark peak  $k$ , the matching window defines how far an ensemble detection can lie and still be attributed to the same fire event:

$$\text{window}_k = \max(\text{mFRI}_z/2, \text{CI}_{95,k}/2)$$

$\text{mFRI}_z$  - mean FRI for the zone containing peak  $k$

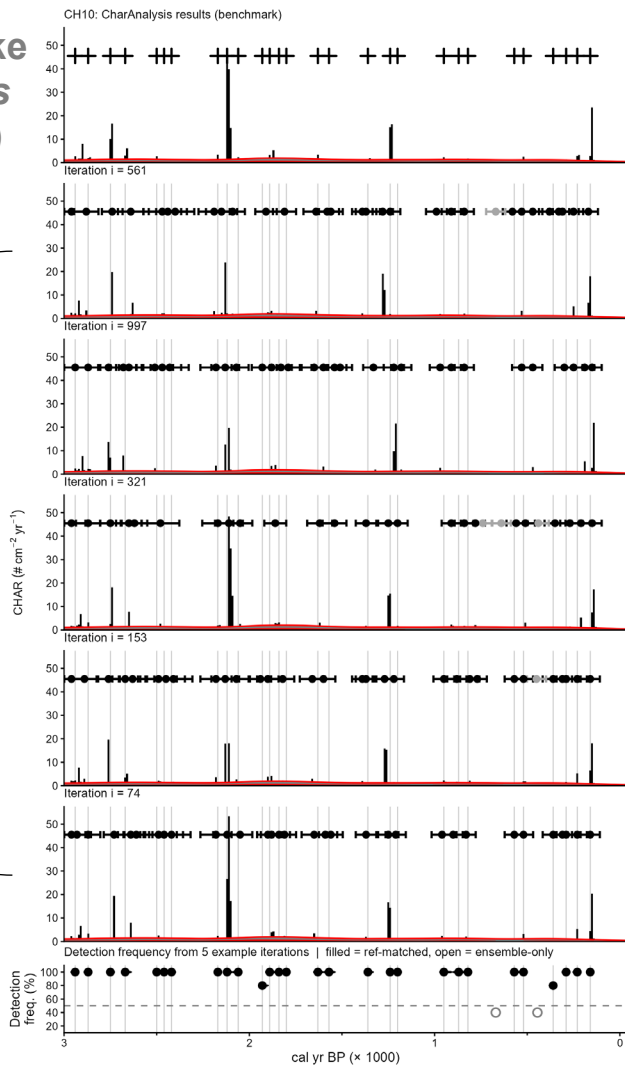
$\text{CI}_{95,k}$  - 95% width of chronological uncertainty at peak  $k$

$\text{mFRI}_z/2$ : once a peak shifts > half the mean FRI, it is considered to be a different event

$\text{CI}_{95,k}/2$ : expands window where chronology is poorly constrained. Ensemble detections of the same peak, shifted by an uncertain chronology, are still correctly attributed

In practice,  $\text{CI}_{95,k}/2$  should define windows in most records;  $\text{mFRI}_z/2$  is a min. bound in well-constrained records (e.g., CH10).

## Chickaree Lake *CharAnalysis* (benchmark)

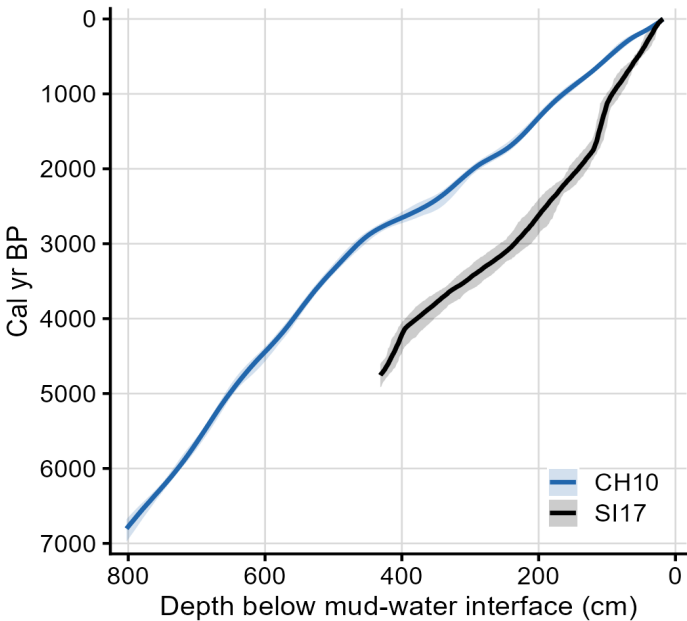


Matched peaks (black)  
Unmatched (white)



# Site examples: CH10 and SI17

|                                 | CH10<br>(Chickaree Lake, CO)                                           | SI17<br>(Silver Lake, MT)                                             |
|---------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Chronological control           | Unusually well-dated<br>(25 $^{14}\text{C}$ dates, $^{210}\text{Pb}$ ) | Fewer dates<br>(15 $^{14}\text{C}$ , 3 tephra, 14 $^{210}\text{Pb}$ ) |
| Age-depth model source          | MCAgeDepth                                                             | Bacon                                                                 |
| Benchmark peaks                 | 59                                                                     | 25                                                                    |
| Mean FRI                        | 104 yr                                                                 | 193 yr                                                                |
| Matching window (median $\pm$ ) | $\pm 72$ yr                                                            | $\pm 162$ yr                                                          |
| Detection freq (median)         | 99%                                                                    | 99%                                                                   |
| Timing SD (median)              | 23 yr                                                                  | 53 yr                                                                 |
| Ensemble-only peaks             | 9 (6 near-ref, 3 indep.)                                               | 0 (coherence filter)                                                  |



Median chronology  $\pm$  95% CI. CH10: narrow, uniform. SI17: wider.





# CH10 — ensemble results

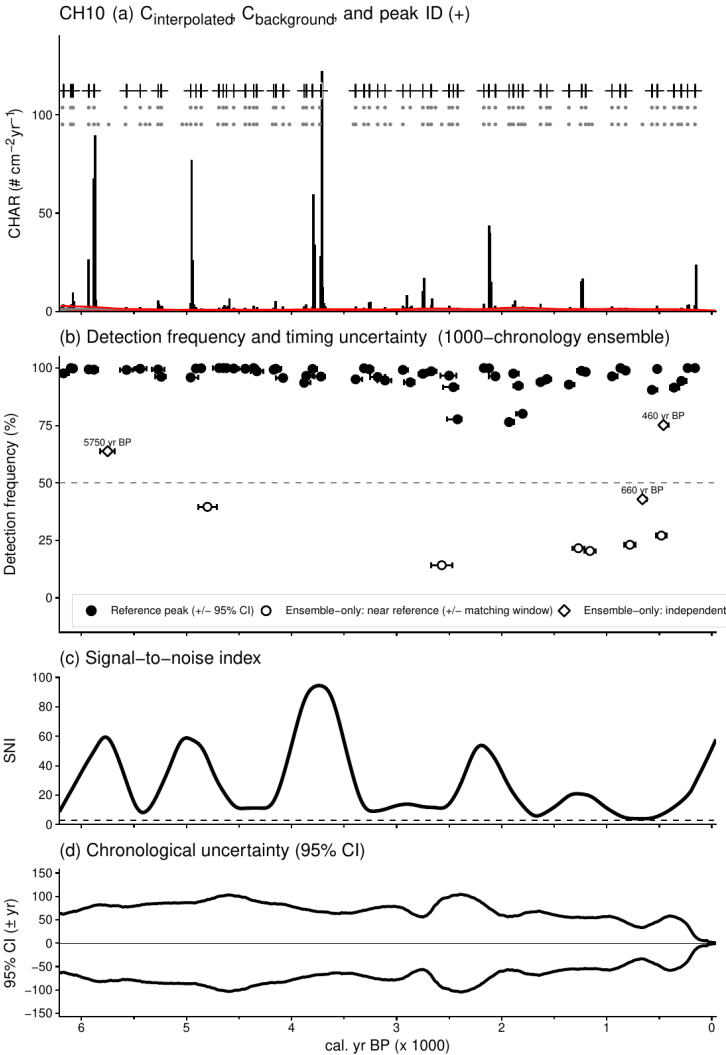
Benchmark:

Detection frequency, with 95% timing CI:

- Reference peaks (black circles)
- Ensemble-only: near reference (white circles)
- Ensemble-only: independent (white diamonds)

SNI:

Age-depth 95% CI:





# SI17 — ensemble results

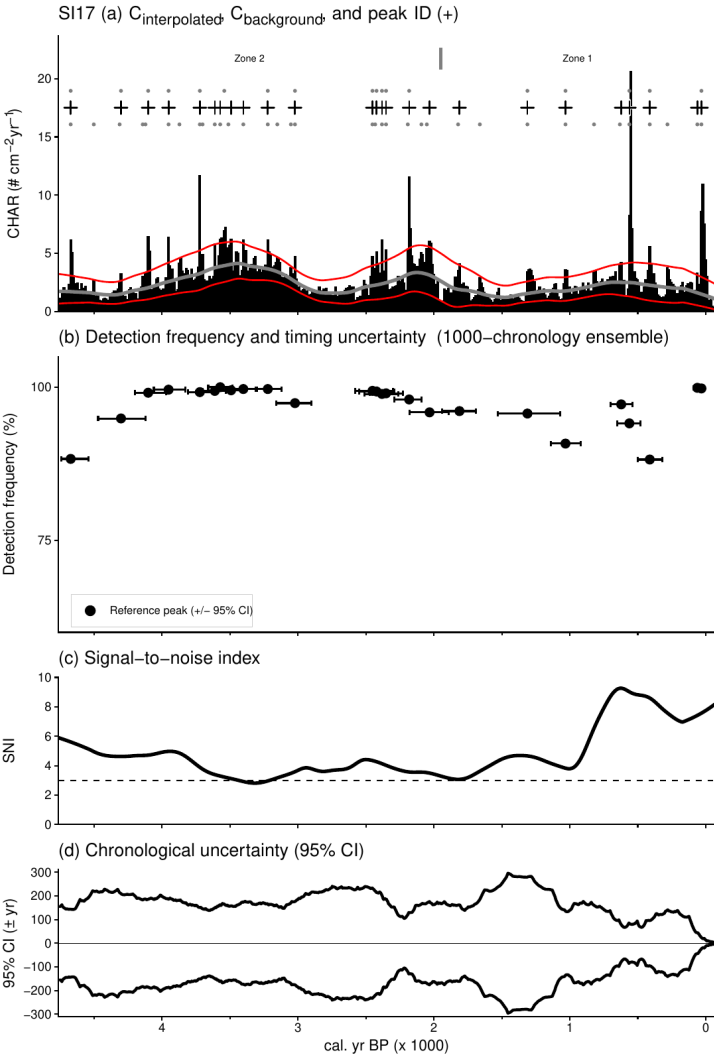
Benchmark:

Detection frequency, with 95% timing CI:

- Reference peaks (black circles)
- Ensemble-only: near reference (NONE)
- Ensemble-only: independent (NONE)

SNI:

Age-depth 95% CI:





# Status & next steps

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## Done

- SNI < 3.0 advisory in console output and added as Step 2 in recommended workflow (vignette)
- Figure reorganization: diagnostic (1–5) vs. analytical (6–9)
- Background sensitivity function added to R, to match Matlab version
- Chronological uncertainty working version

## Up next

- Package `char_run_ensemble()` as exported function with Roxygen docs
- Add rbacon / rplum interface
- Write `chronological_uncertainty.Rmd` vignette
- Apply ensemble to additional sites beyond CH10 and SI17
- Package dev work as v2.0.4 (SNI advisory, figure reorganization, age-uncertainty function)